

V6.2 · Bayesian Confidence Propagation

Uncertainty carried from replicate noise to the theme layer

Objective	Replace a deterministic point estimate at each level with mean, variance, entropy and confidence.
Method	The empirical covariance of an analyte's replicate coordinates is propagated through the two linear maps by the delta method: $\Sigma_{\text{mss}} = M^T \Sigma_{\text{coord}} M$ and $\Sigma_{\text{theme}} = S^T \Sigma_{\text{mss}} S$. Posterior, entropy, margin and confidence are computed from the normalised theme vector.
Frozen	Atlas 09ed804a40836f4a05a91ba10900cded; the V6 MSS map M is an input, never rewritten.
Reproduce	results/v6_rebuild/code/p10 → p11 → p12 → p13

Method

$\Sigma_{\text{mss}} = M^T \Sigma_{\text{coord}} M$
 $\Sigma_{\text{theme}} = S^T \Sigma_{\text{mss}} S$
posterior $p = \text{theme} / \Sigma \text{ theme}$, $H = \text{normalised entropy}(p)$
margin = $p_{(1)} - p_{(2)}$, confidence = $p_{(1)} \cdot (1 - H)$

Part 12 — Bayesian (delta-method) uncertainty propagation from replicate noise

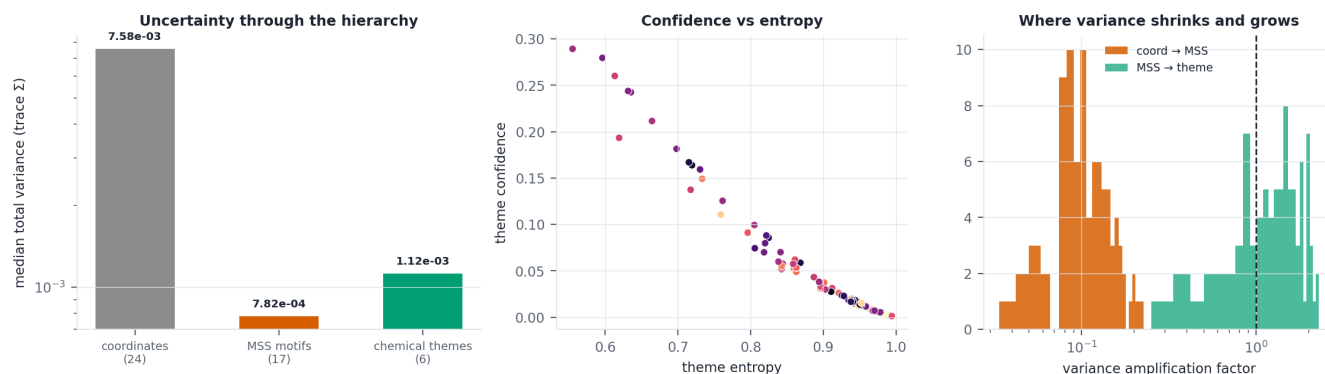


Figure 1 — Total variance at each level, confidence against entropy, and the distribution of variance amplification.

Calibration of the V6.2 soft theme layer

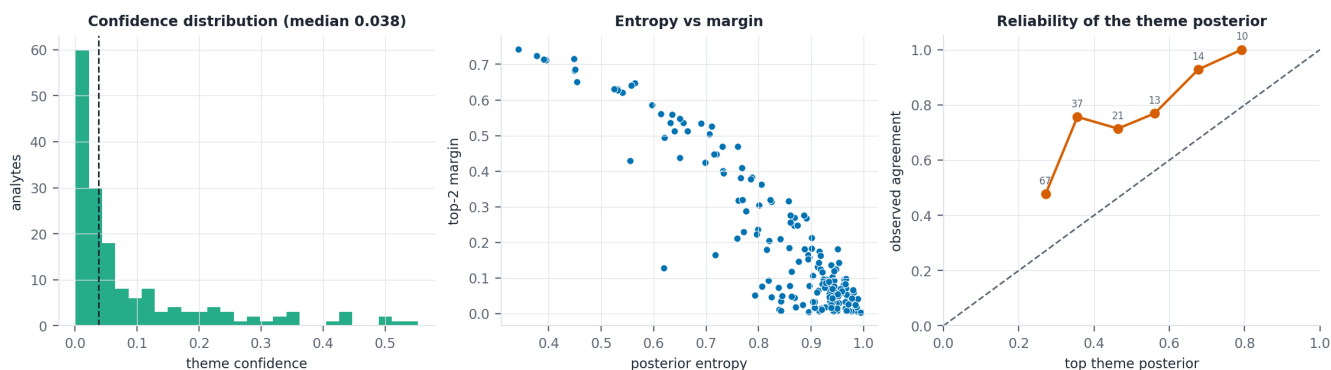


Figure 2 — Confidence distribution, entropy-margin relation and the reliability of the theme posterior.

Results

analytes with ≥ 2 replicates	87
median coord $\rightarrow$ MSS variance ratio	0.1088
median MSS $\rightarrow$ theme variance ratio	1.3912
median theme confidence	0.0235
mean posterior entropy	0.8359
mean top-2 margin	0.2066

The MSS layer denoises. Replicate variance shrinks by roughly 9 $\times$ from the 24 coordinates to the 17 motifs (median ratio 0.109) — pooling components into motifs averages away measurement noise. The theme layer then *amplifies* it slightly (ratio 1.39), because a soft, low-dimensional projection concentrates correlated error. Abstraction is not uniformly stabilising, and the two steps behave differently.

Interpretation

Confidence is low in absolute terms (median 0.024) because it multiplies the top posterior by $(1 - \text{entropy})$ and entropy is high at $K=6$. It should be read comparatively, and the reliability plot shows the posterior is **under-confident**: observed agreement exceeds the stated posterior across most of the range. That is the safe direction to err, but it means the number is not a probability.

Limitations

- The delta method is a first-order approximation; the maps are linear so it is exact for the mean, but the posterior normalisation is not linear.
- Only analytes with ≥ 2 replicates contribute (87 of 167).
- Confidence is a heuristic composite, not a calibrated probability.

Recommendations

- Calibrate the posterior (temperature scaling on the reliability curve) before any downstream consumer treats it as a probability.
- Report Σ_{theme} diagonals alongside the point estimate in the BSV.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. All values recomputed from results/v6_rebuild/.